

EDUCATION

- Northeastern University** Boston, MA
GPA: 3.92/4.0 Class of 2029

PROFESSIONAL EXPERIENCE

- Cerebral Valley Enterprises, Inc.** San Francisco, CA (Hybrid)
Member of Technical Staff (Infrastructure), Part-Time Dec. 2025 - Now
 - Infrastructure:** Migrated our deployment platform to a new Terraform infrastructure-as-code repository to scale with our growing user base of 100,000+ users. Set up an edge cluster platform built on EKS for low-latency compute in APAC regions. Rewrote our real-time WebSocket gateway to support up to 250k events/second.
 - Agents:** Designed & implemented a stateless event-driven agent runtime in TypeScript that enabled horizontal scaling and replayable task execution across agentic workloads. Worked with product teams and close business partners to set up a dog feeding program that improved task completion rate from 35% to 85%.
- Rule Your Own Game, Inc.** Houston, TX (Remote)
Software Engineer (Infrastructure), Part-Time Jan. 2025 - Now
 - Infrastructure:** Deployed highly available infrastructure on our bare-metal hardware using Kubernetes (k3s, Agones), managing 60+ game server instances supporting 1,500+ concurrent players and 3M+ unique players.
 - Content Modernization:** Designed an extensive Kotlin framework enabling rapid content production. Shipped rewrites of 4 major minigames in 3 months, saving thousands in development costs. Built an automated chat moderation system using OpenAI's Omni Moderation model, then revised the system by building an in-house low-latency inference platform in Rust that powers two custom SLMs that help prevent abusive chat on our Minecraft multiplayer network.
 - Team Leadership:** Managed a team of 3 software engineers asynchronously, onboarded and trained 5+ engineers on a modernized codebase through 1-on-1 mentoring sessions. Designed a player queue and load-balancing frameworks backed by a replicated state machine, enabling zero-downtime deployments.
- Fragile, Inc.** San Francisco, CA (Hybrid)
Software Engineer, Part-Time Nov. 2024 - Jan. 2025
 - Infrastructure:** Worked on the core modular monolith TypeScript backend services and AWS cloud infrastructure on ECS that powers Fragile's hardware subscription platform (billing/logistics).
 - Frontend:** Designed and upgraded existing internal tooling used by Fragile's customer service team for enhanced productivity.
- NYU IT High Speed Research Network** Brooklyn, NY (Hybrid)
Software Engineer Dec. 2022 - Dec. 2024
 - Web Development:** Led end-to-end development of a web dashboard built in TypeScript using Next.js, actively used by NYU research scientists. Built a modern version of Corelink's dated API client for AR/VR research purposes.
 - Backend Systems:** Built a low-overhead, self-managed notification service using PostgreSQL's JSON fields for flexible schemas. Developed a device management system with self-registration, complex permission checks, and audit logging, integrated with network switch software for automated firewall management.
 - Technical Leadership:** Conducted multiple rounds of technical interviews to recruit new undergraduate and postgraduate members for the program.

CERTIFICATIONS

- AWS Certified Machine Learning - Specialty** Sep. 2024
- AWS Certified Solutions Architect - Associate** Sep. 2023

OPEN-SOURCE SOFTWARE

- Autopilot Mail:** Self-hosted email server SDK for TypeScript. Pluggable storage, email transports, and webhook handlers for agentic email inboxes.
- Liftgate:** Open-source game server deployment system, auto-scaling system, and service mesh with gRPC.

TECHNICAL SKILLS

- **Languages:** Kotlin, Java, TypeScript, Python, Rust
- **Frameworks:** Spring Framework, Ktor, React, Next.js, Express.js, gRPC
- **Infrastructure:** Kubernetes (Helm, Agones), AWS (Amazon Web Services), Azure, Docker, CI/CD (Jenkins, TeamCity), Nginx, Tailscale, Terraform, Consul, Nomad, Ubuntu, Debian
- **Datastores:** PostgreSQL, MongoDB, Redis, Zookeeper, DynamoDB, RDS, Apache Kafka/Spark/Flink
- **Observability:** VictoriaMetrics, Grafana, Prometheus, DataDog, OpenTelemetry